

Year 7 · Chapter 6 · Geometrical figures: answers (teacher copy)

6.01 Transforming shapes (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: mirror image on the right of the line
2. Ex 1b: moved 5 right and 3 down
3. Ex 2a: turned upside down through O
4. Ex 2b: turned a quarter turn clockwise
5. Ex 3a: mirror image below the line
6. Ex 3b: turned three-quarters clockwise

Easy · Set A

1. reflected to the right of $x = 7$
2. reflected below the line

Easy · Set A2 (quick drill)

1. (5, 2)
2. (3, 5)
3. (2, 0)
4. (7, 3)
5. (0, 4)
6. (8, 0)

Easy · Set B

1. moved 4 left, 1 up
2. moved 4 right, 6 up

Medium · Set C

1. rotated 180°
2. rotated 90° clockwise

Challenging · Set D

1. rotated 270° clockwise
2. reflected in the diagonal

Skill drill 1 (Easy basics)

- | | | |
|--------------|--------------|--------------|
| 1. (-2, -9) | 15. (-1, -7) | 29. (2, 3) |
| 2. (3, -7) | 16. (1, -7) | 30. (7, -6) |
| 3. (10, -2) | 17. (0, -1) | 31. (8, -4) |
| 4. (0, -4) | 18. (-7, -4) | 32. (-3, -1) |
| 5. (8, -6) | 19. (-7, -2) | 33. (-5, -8) |
| 6. (-1, -1) | 20. (-4, -8) | 34. (-4, -1) |
| 7. (2, 7) | 21. (2, -2) | 35. (3, 2) |
| 8. (-1, 5) | 22. (-2, 5) | 36. (-1, 2) |
| 9. (-6, 0) | 23. (-4, -6) | 37. (6, 2) |
| 10. (0, 3) | 24. (-6, -7) | 38. (6, 6) |
| 11. (1, -1) | 25. (-2, -6) | 39. (7, 4) |
| 12. (-3, -3) | 26. (-2, -1) | 40. (6, 4) |
| 13. (2, 2) | 27. (-6, -2) | |
| 14. (-9, -7) | 28. (8, -6) | |

Skill drill 2 (Medium basics)

- | | | |
|--------------|-------------------------------------|------------------------------|
| 1. (4, -1) | 15. (3, 6) | 22. reflection in the x-axis |
| 2. (-5, 2) | 16. translation 3 right | 23. translation 3 right |
| 3. (2, -6) | 17. reflection in the y-axis | 24. translation 3 right |
| 4. (-4, -5) | 18. rotation of 180° about O | 25. reflection in the y-axis |
| 5. (-6, 6) | 19. rotation of 180° about O | 26. reflection in the y-axis |
| 6. (-4, 3) | 20. rotation of 180° about O | 27. reflection in the y-axis |
| 7. (5, 5) | 21. translation 3 right | 28. reflection in the x-axis |
| 8. (6, 3) | | |
| 9. (-4, 4) | | |
| 10. (4, -2) | | |
| 11. (-4, -1) | | |
| 12. (6, 4) | | |
| 13. (-2, -5) | | |
| 14. (6, 1) | | |

29. reflection in the x-axis

30. reflection in the y-axis

Extension

1. a rotation of 180° about the point where the lines cross
2. Yes: both end in the same position

Exit ticket

1. Level 1: reflection
2. Level 2: (6, 2)
3. Level 3: it points right

6.02 Combined transformations (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: translated, then flipped over the line
2. Ex 1b: rotated, then moved 3 left
3. Ex 2a: reflected, then moved down 1
4. Ex 2b: Often yes: the images end in different places
5. Ex 3a: three images drawn in order
6. Ex 3b: e.g. rotate 180° , then translate

Easy · Set A

1. moved then flipped
2. flipped then moved up

Easy · Set A2 (quick drill)

1. reflection
2. translation
3. rotation
4. reflection
5. translation
6. rotation

Easy · Set B

1. turned then moved down
2. moved then turned

Medium · Set C

1. three images
2. three images

Challenging · Set D

1. e.g. rotate 180° then translate
2. reflect in a vertical line; or reflect then translate
3. Not always: e.g. a reflection in a horizontal line and a translation to the right give the same, but a translation up does not
4. translations

Skill drill 1 (Easy basics)

- | | | |
|-----------------|-----------------|--------------|
| 1. reflection | 11. reflection | 21. 2 right |
| 2. reflection | 12. translation | 22. 6 right |
| 3. translation | 13. translation | 23. 5 right |
| 4. rotation | 14. translation | 24. 11 right |
| 5. rotation | 15. 2 right | 25. 8 left |
| 6. rotation | 16. 12 left | 26. 7 left |
| 7. rotation | 17. 10 right | 27. 1 right |
| 8. reflection | 18. 4 right | 28. 6 left |
| 9. rotation | 19. 14 right | 29. 12 left |
| 10. translation | 20. 4 left | |

Skill drill 2 (Medium basics)

- | | | |
|------------|------------|--------------|
| 1. (4, -5) | 5. (4, -1) | 9. (0, 5) |
| 2. (8, -3) | 6. (-2, 5) | 10. (-2, -4) |
| 3. (4, 5) | 7. (7, -5) | 11. (-5, -2) |
| 4. (-2, 4) | 8. (-4, 5) | 12. (-5, 3) |

- | | | |
|---------------------|--------------------|--------------------|
| 13. 6 left, 1 down | 17. 1 right, 1 up | 21. 1 right, 1 up |
| 14. 1 left, 3 down | 18. 5 right, 4 up | 22. 4 right, 5 up |
| 15. 5 right, 3 up | 19. 4 left, 3 up | 23. 5 left, 4 down |
| 16. 6 right, 1 down | 20. 2 left, 2 down | 24. 1 right, 5 up |

Extension

1. $(-1, -5)$
2. $(-4, -1)$

Exit ticket

1. Level 1: the image after one transformation
2. Level 2: $(4, 6)$
3. Level 3: $(0, -3)$

6.03 Line symmetry (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: 2
2. Ex 1b: 4
3. Ex 2a: 6
4. Ex 2b: 0
5. Ex 3a: 2
6. Ex 3b: infinitely many

Easy · Set A

1. 3
2. 1
3. 2
4. 1
5. 1
6. 0

Easy · Set A2 (quick drill)

1. 4
2. 2
3. 3
4. 1
5. 6
6. 0
7. 1
8. infinitely many
9. 5

Easy · Set B

1. 5
2. 4
3. 1

Medium · Set C

1. 8
2. 0
3. 0
4. 1
5. 0
6. 2

Challenging · Set D

1. 10
2. e.g. A, B, C, D, E, M, T, U, V, W, Y
3. e.g. a rectangle
4. No: 2 axes would make all sides equal, giving 3

Skill drill 1 (Easy basics)

- | | | |
|------|-------|---------------------|
| 1. 1 | 7. 0 | 13. 2 |
| 2. 0 | 8. 4 | 14. infinitely many |
| 3. 1 | 9. 3 | 15. 0 |
| 4. 1 | 10. 5 | 16. 1 |
| 5. 6 | 11. 8 | 17. 2 |
| 6. 1 | 12. 2 | 18. 0 |

- | | | |
|-------|-------|-------|
| 19. 2 | 25. 0 | 31. 2 |
| 20. 1 | 26. 1 | 32. 1 |
| 21. 1 | 27. 0 | 33. 0 |
| 22. 1 | 28. 1 | 34. 1 |
| 23. 1 | 29. 1 | |
| 24. 2 | 30. 1 | |

Skill drill 2 (Medium basics)

- | | | |
|-------|-----------|-----------|
| 1. 14 | 11. 19 | 21. false |
| 2. 6 | 12. 15 | 22. true |
| 3. 12 | 13. 5 | 23. true |
| 4. 20 | 14. 10 | 24. false |
| 5. 16 | 15. 7 | 25. false |
| 6. 17 | 16. 9 | 26. true |
| 7. 8 | 17. true | 27. true |
| 8. 13 | 18. true | 28. false |
| 9. 3 | 19. false | |
| 10. 4 | 20. true | |

Extension

1. a shape with 2 axes
2. 12

Exit ticket

1. Level 1: 3
2. Level 2: 1
3. Level 3: 6

6.04 Rotational symmetry (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 2
2. Ex 1b: 4
3. Ex 2a: 2
4. Ex 2b: none (order 1)
5. Ex 3a: 2
6. Ex 3b: 72°

Easy · Set A

1. order 3
2. order 2
3. none
4. order 6
5. none
6. order 4

Easy · Set A2 (quick drill)

1. 4
2. 2
3. 3
4. 1
5. 6
6. 2
7. 2
8. 1
9. 8

Easy · Set B

1. 2
2. 5
3. none

Medium · Set C

1. 8, 45°
2. 4, 90°
3. 2, 180°
4. 2, 180°
5. 3, 120°
6. none

Challenging · Set D

1. e.g. H, I, N, O, S, X, Z
2. Yes, e.g. a parallelogram or the letter S
3. 5
4. e.g. a square or a plus sign

Skill drill 1 (Easy basics)

- | | | |
|-------------|-------|-------|
| 1. 2 | 13. 1 | 25. 2 |
| 2. 4 | 14. 6 | 26. 1 |
| 3. infinite | 15. 1 | 27. 2 |
| 4. 2 | 16. 2 | 28. 1 |
| 5. 1 | 17. 2 | 29. 2 |
| 6. 5 | 18. 1 | 30. 1 |
| 7. 1 | 19. 1 | 31. 1 |
| 8. 1 | 20. 2 | 32. 2 |
| 9. 2 | 21. 1 | 33. 1 |
| 10. 1 | 22. 1 | 34. 1 |
| 11. 8 | 23. 1 | |
| 12. 3 | 24. 1 | |

Skill drill 2 (Medium basics)

- | | | |
|------------------|-------------------|-----------|
| 1. order 9, 40° | 10. order 20, 18° | 19. false |
| 2. order 3, 120° | 11. order 15, 24° | 20. true |
| 3. order 36, 10° | 12. order 45, 8° | 21. true |
| 4. order 10, 36° | 13. order 5, 72° | 22. true |
| 5. order 40, 9° | 14. order 4, 90° | 23. false |
| 6. order 18, 20° | 15. order 12, 30° | 24. true |
| 7. order 6, 60° | 16. true | 25. true |
| 8. order 8, 45° | 17. true | 26. false |
| 9. order 24, 15° | 18. true | 27. false |

Extension

1. 12
2. order 6: 60° is larger than 45°

Exit ticket

1. Level 1: 4
2. Level 2: no
3. Level 3: 60°

6.05 Classifying triangles (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: equilateral, acute-angled
2. Ex 1b: scalene, right-angled
3. Ex 2a: isosceles, acute-angled; $\angle F = \angle H$
4. Ex 2b: scalene, obtuse-angled
5. Ex 3a: scalene, acute-angled
6. Ex 3b: equilateral, acute-angled

Easy · Set A

1. isosceles
2. scalene
3. equilateral
4. isosceles
5. scalene
6. isosceles

Easy · Set A2 (quick drill)

1. acute
2. right
3. obtuse
4. acute
5. obtuse
6. right

Easy · Set B

1. right-angled
2. acute-angled

3. obtuse-angled

Medium · Set C

1. isosceles, right-angled
2. isosceles, obtuse-angled
3. equilateral, acute-angled
4. scalene, right-angled
5. scalene, acute-angled
6. isosceles, acute-angled

Challenging · Set D

1. scalene, acute-angled
2. Yes: isosceles right-angled (the 7 cm side is the longest)
3. isosceles, obtuse-angled
4. No: equilateral angles are all 60°

Skill drill 1 (Easy basics)

- | | | |
|------------|-----------------|-----------------|
| 1. obtuse | 15. obtuse | 29. isosceles |
| 2. acute | 16. obtuse | 30. scalene |
| 3. obtuse | 17. obtuse | 31. equilateral |
| 4. acute | 18. acute | 32. isosceles |
| 5. acute | 19. obtuse | 33. isosceles |
| 6. obtuse | 20. acute | 34. equilateral |
| 7. acute | 21. acute | 35. equilateral |
| 8. acute | 22. obtuse | 36. isosceles |
| 9. acute | 23. obtuse | 37. isosceles |
| 10. obtuse | 24. acute | 38. isosceles |
| 11. acute | 25. scalene | 39. scalene |
| 12. acute | 26. equilateral | 40. scalene |
| 13. obtuse | 27. isosceles | |
| 14. obtuse | 28. scalene | |

Skill drill 2 (Medium basics)

- | | | |
|-----------------------|----------------------|---------|
| 1. isosceles, acute | 10. scalene, obtuse | 19. no |
| 2. equilateral, acute | 11. scalene, obtuse | 20. no |
| 3. isosceles, acute | 12. scalene, obtuse | 21. yes |
| 4. isosceles, right | 13. scalene, obtuse | 22. yes |
| 5. scalene, obtuse | 14. scalene, obtuse | 23. yes |
| 6. isosceles, acute | 15. isosceles, acute | 24. yes |
| 7. scalene, obtuse | 16. yes | 25. no |
| 8. isosceles, acute | 17. yes | 26. no |
| 9. isosceles, acute | 18. no | 27. yes |

Extension

1. 40° and 40°
2. equilateral

Exit ticket

1. Level 1: equilateral
2. Level 2: isosceles, right-angled
3. Level 3: isosceles, obtuse-angled

6.06 Angle sum of a triangle (booklet p. 43)

WE DO examples (teacher)

1. Ex 1a: $x = 59$
2. Ex 1b: $u = 51$
3. Ex 2a: $a = 127$
4. Ex 2b: $y = 50, z = 80$
5. Ex 3a: $n = 62, m = 56$
6. Ex 3b: $\angle PQR = 36^\circ$ (C)

Easy · Set A

1. $x = 59$
2. $v = 76$
3. $k = 50$
4. $d = 85$
5. $u = 51$

6. $p = 55$

Easy · Set A2 (quick drill)

1. 70°
2. 55°
3. 90°
4. 50°
5. 36°
6. 40°

Easy · Set B

1. $a = 127$
2. $p = 100$
3. $w = 27$

Medium · Set C

1. $y = 50, z = 80$
2. $n = 62, m = 56$
3. $t = 60$
4. $b = 35$
5. $c = 45$
6. $q = 36$

Challenging · Set D

1. $x = 30$
2. 72° each
3. No: $100 + 90$ is already more than 180
4. $53\frac{1}{3}^\circ, 53\frac{1}{3}^\circ, 73\frac{1}{3}^\circ$

Skill drill 1 (Easy basics)

- | | | |
|----------------|-----------------|----------------|
| 1. 16° | 15. 101° | 29. 52° |
| 2. 75° | 16. 25° | 30. 51° |
| 3. 84° | 17. 26° | 31. 80° |
| 4. 62° | 18. 55° | 32. 26° |
| 5. 129° | 19. 23° | 33. 29° |
| 6. 25° | 20. 25° | 34. 33° |
| 7. 133° | 21. 22° | 35. 15° |
| 8. 37° | 22. 77° | 36. 43° |
| 9. 42° | 23. 30° | 37. 62° |
| 10. 79° | 24. 75° | 38. 23° |
| 11. 39° | 25. 79° | 39. 67° |
| 12. 53° | 26. 72° | 40. 42° |
| 13. 91° | 27. 12° | |
| 14. 48° | 28. 16° | |

Skill drill 2 (Medium basics)

- | | | |
|--|---|-------------------------------|
| 1. 78° and 78° | 9. apex 102° ,
other base 39° | 16. 49° and 49° |
| 2. apex 30° , other
base 75° | 10. apex 72° , other
base 54° | 17. $x = 21$ |
| 3. apex 70° , other
base 55° | 11. apex 138° ,
other base 21° | 18. $x = 48$ |
| 4. 65° and 65° | 12. apex 36° , other
base 72° | 19. $x = 35$ |
| 5. apex 130° ,
other base 25° | 13. apex 104° ,
other base 38° | 20. $x = 16$ |
| 6. 15° and 15° | 14. apex 22° , other
base 79° | 21. $x = 47$ |
| 7. apex 140° ,
other base 20° | 15. apex 88° , other
base 46° | 22. $x = 47$ |
| 8. apex 118° ,
other base 31° | | 23. $x = 27$ |
| | | 24. $x = 15$ |
| | | 25. $x = 25$ |
| | | 26. $x = 64$ |
| | | 27. $x = 33$ |
| | | 28. $x = 19$ |

Extension

1. they form a straight line (180°)
2. two obtuse angles already add to more than 180°

Exit ticket

1. Level 1: 70°
2. Level 2: 62°
3. Level 3: 46°

6.07 Exterior angle of a triangle (booklet p.

51)

WE DO examples (teacher)

1. Ex 1a: $x = 115$
2. Ex 1b: $x = 111$
3. Ex 2a: $x = 112$
4. Ex 2b: $p = 96$
5. Ex 3a: $m = 99, n = 81$
6. Ex 3b: it is the sum of both, so it is bigger than either

Easy · Set A

1. $x = 115$
2. $x = 111$
3. $a = 110$
4. $b = 120$
5. $c = 120$
6. $d = 90$

Easy · Set A2 (quick drill)

1. 110°
2. 120°
3. 120°
4. 120°
5. 70°
6. 115°

Easy · Set B

1. $x = 70$
2. $p = 96$
3. $y = 55$

Medium · Set C

1. $m = 99, n = 81$
2. $j = 108, k = 72$
3. $r = 55, s = 70$
4. $t = 60$
5. $u = 62, v = 90$
6. $w = 60$

Challenging · Set D

1. $x = 37$
2. $70^\circ, 70^\circ, 40^\circ$
3. 360°
4. Yes, if the adjacent interior angle is obtuse

Skill drill 1 (Easy basics)

- | | | |
|-----------------|-----------------|-----------------|
| 1. 94° | 15. 109° | 29. 62° |
| 2. 124° | 16. 135° | 30. 102° |
| 3. 87° | 17. 154° | 31. 131° |
| 4. 139° | 18. 149° | 32. 87° |
| 5. 94° | 19. 120° | 33. 52° |
| 6. 72° | 20. 41° | 34. 114° |
| 7. 88° | 21. 64° | 35. 125° |
| 8. 89° | 22. 113° | 36. 56° |
| 9. 80° | 23. 147° | 37. 160° |
| 10. 139° | 24. 156° | 38. 139° |
| 11. 104° | 25. 37° | 39. 158° |
| 12. 134° | 26. 34° | 40. 136° |
| 13. 143° | 27. 78° | |
| 14. 133° | 28. 105° | |

Skill drill 2 (Medium basics)

- | | | |
|---------------|----------------|-----------------|
| 1. 81° | 6. 17° | 11. 143° |
| 2. 93° | 7. 56° | 12. 84° |
| 3. 92° | 8. 54° | 13. 76° |
| 4. 33° | 9. 11° | 14. 130° |
| 5. 25° | 10. 13° | 15. 82° |

- | | | |
|----------------|--------------|--------------|
| 16. 25° | 21. $x = 58$ | 26. $x = 20$ |
| 17. $x = 17$ | 22. $x = 70$ | 27. $x = 43$ |
| 18. $x = 79$ | 23. $x = 66$ | 28. $x = 75$ |
| 19. $x = 57$ | 24. $x = 25$ | |
| 20. $x = 36$ | 25. $x = 37$ | |

Extension

1. 144°
2. 55° and 75°

Exit ticket

1. Level 1: 115°
2. Level 2: 65°
3. Level 3: 40°

6.08 Quadrilaterals (booklet p. 59)

WE DO examples (teacher)

1. Ex 1a: non-convex
2. Ex 1b: parallelogram
3. Ex 2a: rhombus
4. Ex 2b: trapezium
5. Ex 3a: a parallelogram with all sides equal
6. Ex 3b: trapezium

Easy · Set A

1. convex
2. non-convex
3. convex
4. non-convex
5. convex
6. non-convex

Easy · Set A2 (quick drill)

1. square
2. parallelogram
3. trapezium
4. rhombus
5. rectangle
6. kite

Easy · Set B

1. rectangle
2. kite
3. square

Medium · Set C

1. 2 axes, order 2
2. 1 axis, none
3. 4 axes, order 4
4. 0 axes, order 2
5. 2 axes, order 2
6. 1 axis, none

Challenging · Set D

1. parallelogram
2. a parallelogram with a right angle
3. Yes: it is a parallelogram with all sides equal
4. Yes: both pairs of opposite sides are parallel

Skill drill 1 (Easy basics)

- | | | |
|------------------------|------------------|-----------|
| 1. kite | 8. kite | 16. true |
| 2. rhombus | 9. parallelogram | 17. false |
| 3. rectangle | 10. rhombus | 18. false |
| 4. isosceles trapezium | 11. trapezium | 19. false |
| 5. parallelogram | 12. rectangle | 20. false |
| 6. square | 13. true | 21. true |
| 7. square | 14. true | 22. false |
| | 15. true | |

Skill drill 2 (Medium basics)

- | | | |
|--|-------------------|-------------------------|
| 1. 1 pair, 0 | 6. 0 pairs, 0 | 12. rhombus |
| 2. 2 pairs, 4 | 7. 2 pairs, 0 | 13. rhombus |
| 3. 2 pairs, 4 | 8. kite | 14. isosceles trapezium |
| 4. 2 pairs, 0 | 9. rectangle | 15. square |
| 5. 1 pair, 0 (a right trapezium has 2) | 10. parallelogram | |
| | 11. square | |

Extension

1. a kite is separate; yes, a rhombus is a kite
2. rectangle and rhombus

Exit ticket

1. Level 1: yes
2. Level 2: trapezium
3. Level 3: a quadrilateral with both pairs of opposite sides parallel

6.09 Angle sum of a quadrilateral (booklet p. 67)

WE DO examples (teacher)

1. Ex 1a: $y = 70$
2. Ex 1b: $x = 81$
3. Ex 2a: $m = 26$
4. Ex 2b: $y = 59$
5. Ex 3a: a diagonal splits it into 2 triangles: $2 \times 180^\circ$
6. Ex 3b: 90° each

Easy · Set A

1. $y = 70$
2. $x = 81$
3. $m = 26$
4. $a = 105$
5. $b = 95$
6. $c = 55$

Easy · Set A2 (quick drill)

1. 90°
2. 80°
3. 85°
4. 120°
5. 90°
6. 90°

Easy · Set B

1. $y = 59$
2. $d = 63$, $e = 117$
3. $f = 120$

Medium · Set C

1. $x = 53\frac{1}{3}$
2. $k = 60$
3. $p = 110$, $q = 70$, $r = 110$
4. $t = 95$
5. $u = 87$
6. $w = 36$

Challenging · Set D

1. 36° , 72° , 108° , 144°
2. No: $4 \times$ more than 90° is more than 360°
3. 122° , 58° , 122°
4. 90° each

Skill drill 1 (Easy basics)

- | | | |
|---------------|----------------|----------------|
| 1. 54° | 4. 59° | 7. 80° |
| 2. 40° | 5. 106° | 8. 56° |
| 3. 76° | 6. 63° | 9. 144° |

- | | | |
|-----------------|--------------------------------------|--------------------------------------|
| 10. 113° | 19. 88° | 28. $138^\circ, 42^\circ, 138^\circ$ |
| 11. 56° | 20. 107° | 29. $61^\circ, 119^\circ, 61^\circ$ |
| 12. 34° | 21. 72° | 30. $157^\circ, 23^\circ, 157^\circ$ |
| 13. 84° | 22. 142° | 31. $149^\circ, 31^\circ, 149^\circ$ |
| 14. 81° | 23. 54° | 32. $34^\circ, 146^\circ, 34^\circ$ |
| 15. 55° | 24. 63° | 33. $153^\circ, 27^\circ, 153^\circ$ |
| 16. 86° | 25. $69^\circ, 111^\circ, 69^\circ$ | 34. $144^\circ, 36^\circ, 144^\circ$ |
| 17. 144° | 26. $154^\circ, 26^\circ, 154^\circ$ | 35. $74^\circ, 106^\circ, 74^\circ$ |
| 18. 59° | 27. $68^\circ, 112^\circ, 68^\circ$ | 36. $20^\circ, 160^\circ, 20^\circ$ |

- rhombus, square, kite
- trapezium
- rhombus, square

Medium · Set C

- bisect each other
- equal, bisect
- bisect at 90°
- equal, bisect at 90°
- one bisects the other at 90°
- none of these (equal if isosceles)

Challenging · Set D

- $CD = 7$ cm, $AD = 4$ cm, $\angle C = 65^\circ$
- 90°
- square
- kite

Skill drill 1 (Easy basics)

- | | | |
|-----------|---|---|
| 1. true | 12. false | 19. square, rhombus |
| 2. true | 13. true | 20. square, rectangle, isosceles trapezium |
| 3. true | 14. true | 21. square, rectangle, rhombus, parallelogram |
| 4. false | 15. true | 22. trapezium |
| 5. false | 16. true | |
| 6. true | 17. square, rhombus, kite | |
| 7. true | 18. square, rectangle, rhombus, parallelogram | |
| 8. false | | |
| 9. true | | |
| 10. false | | |
| 11. true | | |

Skill drill 2 (Medium basics)

- | | | |
|----------------|---|--|
| 1. 50° | 11. 12 cm | 16. it has two pairs of equal adjacent sides |
| 2. 28 cm | 12. 9 cm | 17. it has 4 right angles and opposite sides equal |
| 3. 100° | 13. it has 4 equal sides | 18. it has two pairs of equal adjacent sides |
| 4. 65° | 14. both pairs of opposite sides are parallel | |
| 5. 13 cm | 15. both pairs of opposite sides are parallel | |
| 6. 110° | | |
| 7. 12 cm | | |
| 8. 90° | | |
| 9. 125° | | |
| 10. 28 cm | | |

Extension

- 5 cm
- the diagonal bisects each 90° angle

Exit ticket

- Level 1: rectangle or square
- Level 2: they bisect each other at right angles
- Level 3: rectangle (or square)

Skill drill 2 (Medium basics)

- | | | |
|--------------|---------------|--------------|
| 1. $x = 97$ | 9. $x = 100$ | 17. $x = 51$ |
| 2. $x = 128$ | 10. $x = 129$ | 18. $x = 23$ |
| 3. $x = 103$ | 11. $x = 114$ | 19. $x = 15$ |
| 4. $x = 109$ | 12. $x = 96$ | 20. $x = 27$ |
| 5. $x = 85$ | 13. $x = 50$ | 21. $x = 29$ |
| 6. $x = 77$ | 14. $x = 24$ | 22. $x = 40$ |
| 7. $x = 66$ | 15. $x = 53$ | 23. $x = 36$ |
| 8. $x = 76$ | 16. $x = 10$ | 24. $x = 79$ |

Extension

- 720°
- 108°

Exit ticket

- Level 1: 110°
- Level 2: 80°
- Level 3: $115^\circ, 65^\circ, 115^\circ$

6.10 Properties of quadrilaterals (booklet p.

75)

WE DO examples (teacher)

- Ex 1a: e.g. opposite sides equal; opposite angles equal; diagonals bisect each other
- Ex 1b: C
- Ex 2a: rectangle, square
- Ex 2b: rectangle, square
- Ex 3a: parallelogram, rectangle, rhombus, square
- Ex 3b: parallelogram, rectangle, rhombus, square

Easy · Set A (quick drill)

- | | | |
|----------|----------|-----------|
| 1. true | 5. true | 9. true |
| 2. false | 6. false | 10. false |
| 3. true | 7. true | 11. true |
| 4. false | 8. false | 12. false |

Easy · Set B

- rhombus, square
- rectangle, square
- parallelogram, rhombus, rectangle, square